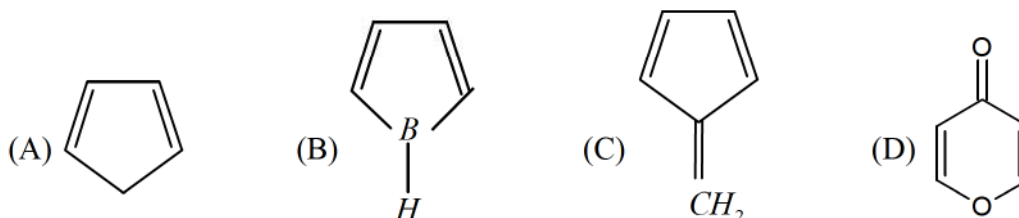


[SINGLE CORRECT CHOICE TYPE]

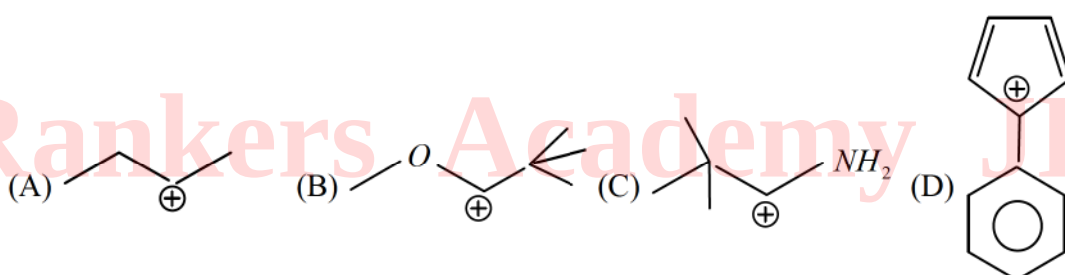
Q.1 Choose the anti-aromatic compound



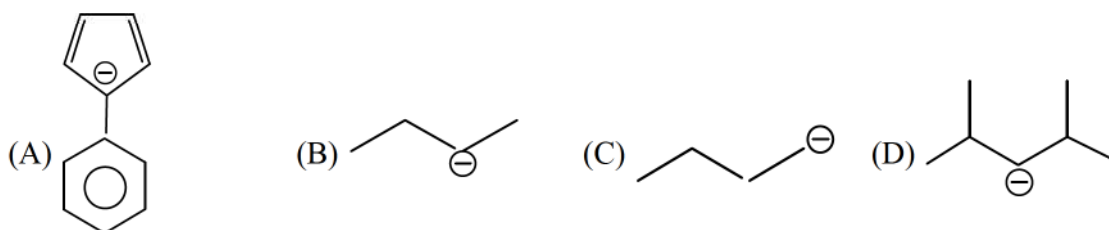
Q.2 Which of the following show addition reaction more readily than substitution reaction when compared to others?

- (A) Borazole (B) Benzene (C) Pyridine (D) Phenol

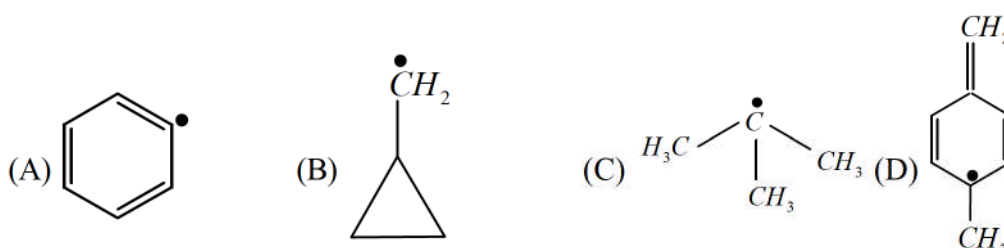
Q.3 Most stable carbocation is?



Q.4 Which of the carbanions is least stable?



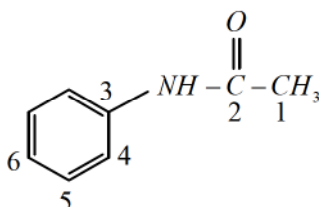
Q.5 Choose the free radical, which is more stable than benzyl radical?



Q.6 Which among the following won't exist (least stable)?



Q.7 Choose the position at which an electrophile may attack



- (A) 3 (B) 6 (C) 2 (D) 5

Q.8 12.5 g of metal carbonate on decomposition produced 7g of metal oxide. Equivalent weight of metal is?

- (A) 20 (B) 30 (C) 40 (D) 50

Q.9 Choose the correct statement?

- (A) D-glucose and L- fructose are enantiomers
(B) cis – 2 – butene and 1 – butene are diastereomers
(C) CO is anhydride of HCOOH
(D) – NHCOCH₃ is stronger +R group than – OCOCH₃

Q.10 Choose the incorrect statement?

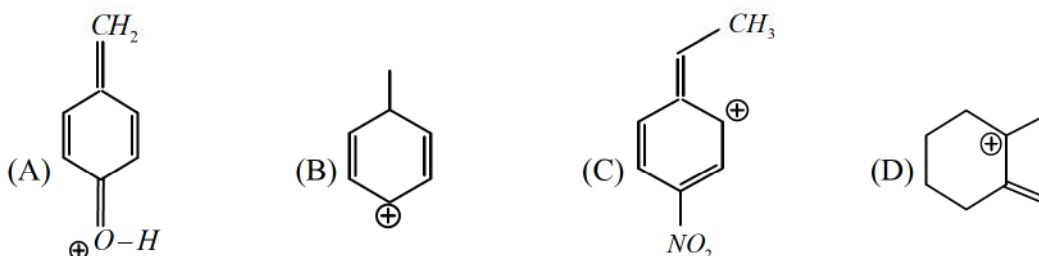
- (A) cyclobutadiene has 2 different C – C bond lengths
(B) no. of planes of symmetries in cyclohexane: chair > Boat > Twist boat form
(C) ethyl carbocation undergoes disproportionation to give ethane & ethene
(D) no. of oxygens per silicon in cyclic/ring silicones is 1

[PARAGRAPH TYPE]

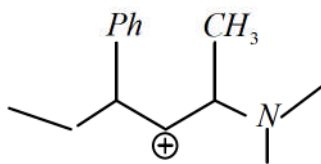
Paragraph for question no. 11 & 12

Many reactions in organic chemistry proceed through carbocation intermediates like etc. In the process they undergo rearrangement to give stable cation before giving product

Q.11 Which of the carbocation undergoes rearrangement to give stable carbocation?



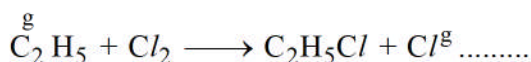
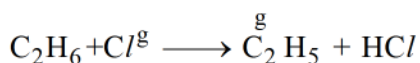
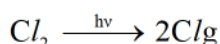
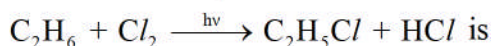
- Q.12 No. of hyper conjugation structures possible for following carbocation after it rearranges to more stable form?



- (A) 2 (B) 4 (C) 5 (D) 3

Paragraph for question no. 13 & 14

Free radicals are the intermediates formed during photohalogenation of alkanes. Mechanism for 1 such common reaction,



And radicals combine together very readily to give some neutral product and E_A (activation energy) for radical combination is ≈ 0

- Q.13 free radicals

- (I) are paramagnetic & neutral
(II) undergo disproportionation
(III) are stabilized by resonance & hyperconjugation
Choose the correct statement(s)

- (A) I & II (B) II & III (C) I & III (D) I, II & III

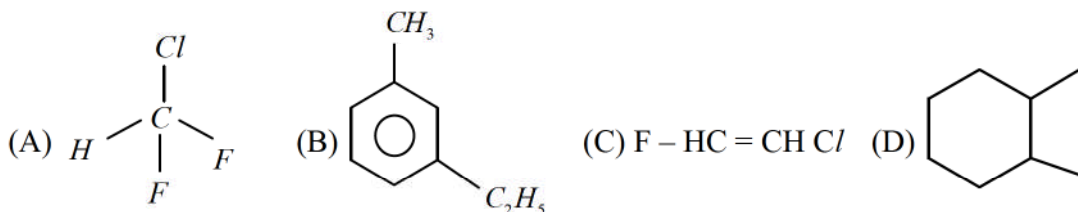
- Q.14 In the reaction & mechanism described in paragraph what are all possible products/side products?

- (I) C_4H_{10} (II) C_2H_4 (III) C_2H_6
(A) only I (B) I & II (C) I & III (D) I, II & III

Paragraph for question no. 15 & 16

Optical activity is the property of asymmetric molecules

- Q.15 Choose the one which is asymmetric?

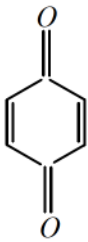
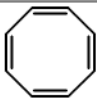


- Q.16 Which of the following has no meso form?

- (A) Tartaric acid (B) Glyceraldehyde
(C) 1, 2 - dichlorocyclopropane (D) 2, 3 - dichlorobutane

[MATCH THE COLUMN]

Q.17

	Column I		Column II
p)		1)	Planar
q)		2)	Non planar
r)		3)	Less stable than corresponding open chain analogue
s)		4)	$\mu \neq 0$

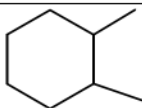
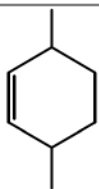
(A) p $\rightarrow$ 1; q $\rightarrow$ 2; r $\rightarrow$ 4; s $\rightarrow$ 3

(B) p $\rightarrow$ 2; q $\rightarrow$ 1; r $\rightarrow$ 3; s $\rightarrow$ 4

(C) p $\rightarrow$ 4; q $\rightarrow$ 3; r $\rightarrow$ 1; s $\rightarrow$ 2

(D) p $\rightarrow$ 3; q $\rightarrow$ 4; r $\rightarrow$ 2; s $\rightarrow$ 1

Q.18

	Column I		Column II
p)	$H_2C=C=CH_2$	1)	Can show geometrical isomerism
q)	$H_3C-CH=C=C=CH_2$	2)	Can show optical isomerism
r)		3)	Has no chiral carbon(s)
s)		4)	Has at least one plane of symmetry

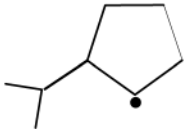
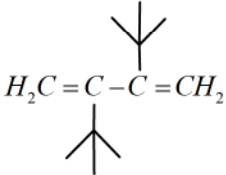
(A) p $\rightarrow$ 4; q $\rightarrow$ 2; r $\rightarrow$ 3; s $\rightarrow$ 1

(B) p $\rightarrow$ 1; q $\rightarrow$ 3; r $\rightarrow$ 2; s $\rightarrow$ 4

(C) p $\rightarrow$ 4; q $\rightarrow$ 3; r $\rightarrow$ 1; s $\rightarrow$ 2

(D) p $\rightarrow$ 2; q $\rightarrow$ 1; r $\rightarrow$ 4; s $\rightarrow$ 3

Q.19

	Column I		Column II
p)	$H_3C-CH=CH-CH=CH_2$	1)	Only 1 type of C-C bond length
q)		2)	No. of hyperconjugation structures is zero
r)		3)	Is stabilized by resonance
s)	$H_2C^{\ominus}-CH=CH_2$	4)	Is stabilized by hyperconjugation

(A) $p \rightarrow 1$; $q \rightarrow 3$; $r \rightarrow 4$; $s \rightarrow 2$ (B) $p \rightarrow 3$; $q \rightarrow 4$; $r \rightarrow 2$; $s \rightarrow 1$ (C) $p \rightarrow 2$; $q \rightarrow 1$; $r \rightarrow 3$; $s \rightarrow 4$ (D) $p \rightarrow 4$; $q \rightarrow 2$; $r \rightarrow 3$; $s \rightarrow 1$

Q.20

	Column I		Column II
p)	O_3	1)	Nucleophile
q)	Br^-	2)	Electrophile
r)	CN^-	3)	Ambident nucleophile
s)	CH_4	4)	Neither nucleophile nor electrophile

(A) $p \rightarrow 1$; $q \rightarrow 1$; $r \rightarrow 3$; $s \rightarrow 4$ (B) $p \rightarrow 3$; $q \rightarrow 1$; $r \rightarrow 4$; $s \rightarrow 2$ (C) $p \rightarrow 4$; $q \rightarrow 2$; $r \rightarrow 1$; $s \rightarrow 3$ (D) $p \rightarrow 2$; $q \rightarrow 1$; $r \rightarrow 3$; $s \rightarrow 4$

ANSWER KEY

Q.1	B	Q.2	A	Q.3	C	Q.4	D	Q.5	D
Q.6	A	Q.7	B	Q.8	A	Q.9	D	Q.10	C
Q.11	B	Q.12	C	Q.13	D	Q.14	D	Q.15	D
Q.16	B	Q.17	A	Q.18	C	Q.19	B	Q.20	D